

circumstances call for even faster evolution of these methods.

More and more companies are rigging operational equipment with sensors to generate a vast amount of data. This data is then analysed using Machine Learning to figure out which areas and processes could use some improvement. This particular method has proven to be tremendously successful in improving supply chain management. Take for example the de-icing fluid incident that occurred at an airport in the US during the peak of winter which resulted in over 55 flights being cancelled as they couldn't be de-iced and prepare for take-off. The company managing the supply of de-icing fluid goofed up big time and they learnt their lesson. Each de-icing truck was fitted with sensors to monitor de-icing fluid levels. This data was then combined with weather data and flight schedules to accurately predict when the airport would run low on de-icing fluid and accordingly the company could schedule a resupply.

Even insurance companies have resorted to using big data analytics to generate user profiles which then allows them to speed up processing of claims by reducing incidences where fraud might be imminent.

Improving Healthcare

The healthcare industry has easy access to a lot of historical data. Big Data analytics are being used to sift through patient medical history to not only gain insight into diseases but to also generate a more specialised treatment for the patients, especially those suffering from chronic illnesses.

Singapore has already implemented a system which takes into account a patient's current condition, lifestyle choices, environment, cultural specifics and other data points to help in speeding up diagnosis. The next step in such a system would be to obtain genetic data and cross reference that with diseases to extrapolate patterns and thus predict when and how a patient might get afflicted with a certain disease. This will also help reduce

hospital expenses for a particular patient as the entire treatment becomes personalised, but also quicker.

Another company, 23andMe, provides at-home DNA testing kit which a customer can use. This kit generates a patient profile and uses data obtained from hundreds of such tests to tell customers if they are at risk for certain diseases.

Who's leading the way?

Aside from the fact that software companies who initiated the big data wave there are some industries which are heavily investing in big data. The energy industry, financial services industry and the healthcare industry



Improve customer relations

are the largest adopters of big data analytics – roughly 66% of management executives from these industries would label their decisions as 'data-driven'.

WHAT'S STOPPING THEM?

Big Data analytics is a new field and that means there exist very few domain experts who can implement data analytics systems with relative ease. And hiring such talent isn't easy given the increasing demand for the same.

It hasn't gone unnoticed that managerial employees view data analytics as a threat since it is the next path of evolution in making decisions. Something which managers do and thus is viewed as a threat to their livelihood.

And lastly, there's the phenomenon of organisational silos. Big Data depends upon voluminous amounts of data from every department of an organisation being analysed and often, there is a close-minded mentality within organisations wherein

employees are more cooperative with members of their own department and a lot less while interacting with those from other departments. This mindset trickles down in the form of different styles and formats being implemented for data in different departments. So when you pool all the data together, you are left with unstructured data rather than coherent data which requires more effort to be effectively analysed using machine learning.

TRYING YOUR HAND AT BIG DATA ANALYTICS

Now that you're convinced about the merits of Big Data analytics, you are probably eager to try your hand at doing some yourself, right? While we did mention that lack of talent was one of the hampering factors towards the greater adoption of Big Data analytics, we should have mentioned that big corporates have highly unstructured data and making sense of that requires experience. So let's scale things down a bit and apply a little bit of Big Data analytics to more simpler data sets. Let's take a survey or perhaps observations pertaining to a scientific study.

We're going to be analysing a sample data set involving an experiment performed in 1979 by M. M. King studying the effect of different diets on rats. Over time the rats developed cancerous tumours so we need to figure out which diet affected the rats the most or the least, i.e. in statistics, this is known as Survival analysis. Speaking of statistics, the following explanation will be a little heavy in statistics terminology so you might as well be prepared to do your research. We will be using IBM SPSS to perform a set of operations on a data set. The data set can be obtained here – <http://dgit.in/Rat-Tumor>. Thankfully, there is a statistical method known as Kaplan-Meier to estimate this very data.

Let's take a look at the data first. We have four columns, the first column is for identification, the second column is diet type, third column indicates the time period for which the rats were cancer-free and the last column indicates if the data was censored. Censoring